04 Accessory organs: liver, gallbladder and pancreas

Team: _____

Basic function (this is all you need): The liver makes bile, the gallbladder stores and concentrates it, and the pancreas delivers enzymes and bicarbonate. All of it empties into the duodenum.

STRUCTURES TO DRAW AND LABEL

- Liver: right hypochondriac and epigastric regions; lobes: right, left, caudate (posterior, by the IVC), quadrate (anterior, by the gallbladder)
- Falciform ligament with the round ligament (remnant of the umbilical vein) in its free edge; porta hepatis (hepatic artery proper, hepatic portal vein, hepatic ducts)
- Bile path: right and left hepatic ducts, common hepatic duct, plus cystic duct forms the bile duct, which joins the main pancreatic duct at the hepatopancreatic ampulla and opens at the major duodenal papilla (hepatopancreatic sphincter)
- Gallbladder: fundus, body, neck; cystic duct with the spiral fold
- Pancreas: retroperitoneal; head in the C of the duodenum with the uncinate process, neck, body, tail at the spleen
- Main pancreatic duct to the major papilla; accessory pancreatic duct to the minor papilla

NERVES AND BLOOD SUPPLY

- Parasympathetic: vagus nerve (CN X) to liver, gallbladder and pancreas
- Sympathetic: greater and lesser splanchnic nerves to the celiac and superior mesenteric ganglia
- Gallbladder contraction is driven mainly by CCK from the duodenum, with vagal input
- These organs are not part of the tube, so they do not have the submucosal and myenteric plexuses. Their nerves travel in plexuses along the hepatic artery and the pancreatic vessels.
- Artery: Liver: hepatic artery proper and hepatic portal vein. Gallbladder: cystic artery. Pancreas: splenic artery branches and the pancreaticoduodenal arteries

THE WALL UNDER THE MICROSCOPE

- Hepatic lobule: hexagon, central vein in the middle, plates of hepatocytes radiating out, sinusoids between the plates
- Portal triad at each corner: branch of hepatic portal vein, branch of hepatic artery, bile duct
- Bile canaliculi between hepatocytes carry bile outward, opposite to blood flow
- Gallbladder: simple columnar epithelium, folded mucosa, no submucosa, muscularis, serosa or adventitia
- Pancreas: serous acini in lobules (about 98 percent of the gland); centroacinar cells inside the acinus; no striated ducts
- Pancreatic islets: pale round clusters scattered among the acini

CELLS AND WHAT THEY MAKE

Hepatocytes	Bile (bile salts, bilirubin, cholesterol), albumin and clotting factors; store glycogen
Kupffer cells (in sinusoids)	Clear worn red cells, debris and microbes
Hepatic stellate (Ito) cells	Store vitamin A
Cholangiocytes (bile duct lining)	Add water and bicarbonate to bile
Gallbladder epithelium	Pull water and salt out of bile to concentrate it
Pancreatic acinar cells	Enzymes, most as inactive precursors: trypsinogen, chymotrypsinogen, procarboxypeptidase; plus amylase, lipase, nucleases
Centroacinar and duct cells	Bicarbonate rich fluid
Islet alpha, beta, delta, PP cells	Glucagon, insulin, somatostatin, pancreatic polypeptide

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Analogy: the lobule is a wheel with two-way traffic

Hub is the central vein, spokes are the hepatocyte plates, and the corners of the rim are the portal triads. Blood drives in from the rim to the hub. Bile drives out from the hub to the rim in the canaliculi. Opposite directions, like two lanes of a road.

Mnemonic: the bile duct adds a C each step

Hepatic ducts join to make the Common hepatic duct. Add the Cystic duct and it becomes the Common bile duct. Pancreas: the head sits in the duodenum's hug and the tail touches the spleen.